

De Montfort University

Data Mining Report

Assignment



Submitted By:

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1. Abstract

Analysing the airline quality dataset we have many features like customer's age, their type and experience in the flight, flight delays and distance, and in-flight services given by the airline. The main target of an airline industry should be the customer satisfaction which was only 33.33% in this dataset. This project addresses this problem of customer dissatisfaction and the cause of it. Further, we explored the dataset for model roles, missing data, standard deviation, data skewness, outliers and correlation. The dataset provided includes categorical and numerical variables. Following the SEMMA framework, we modified the dataset using Data Partition, Filter, Transformation to see different results while building our model that is, Decision Tree and Neural Network in this project. For assessing, we did a Model Comparison, where we found that Decision tree with multiway split is our best performing model. The result presented the overall importance of data directed decision which can improve customer satisfaction, offering valuable perception throughout the airline industry.

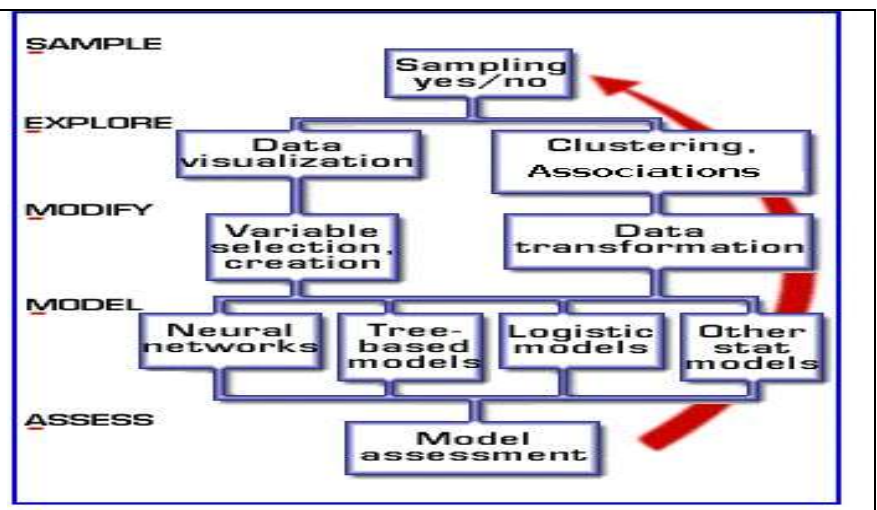
2. Business Problem

Many airline industries face many challenges that can interrupt the operation and leave the customers dissatisfied by the services provided. Problems like: flight delays, inefficient day to day operations can create problems for travelers and they also affect the commercial aspects of the industry. Delay problems can lead to an increase in operational costs as well as leaving the customer unsatisfied can hamper the reputation of the airline.

The aim of this project is to find meaningful information from large amount of data which will enable the airline to focus on improving their passenger's experience. Businesses can understand their client and their responses regarding the services which will help in identifying the patterns and trends. Resolving the issues faced in the past, the airline build strong connection with its customers, ensuring they keep coming back. To mark success in the long run, it is very crucial for businesses to identify the problem using the dataset and act on the information extracted by dataset, which can be done by Data Mining techniques.

3. Methodology - SEMMA framework

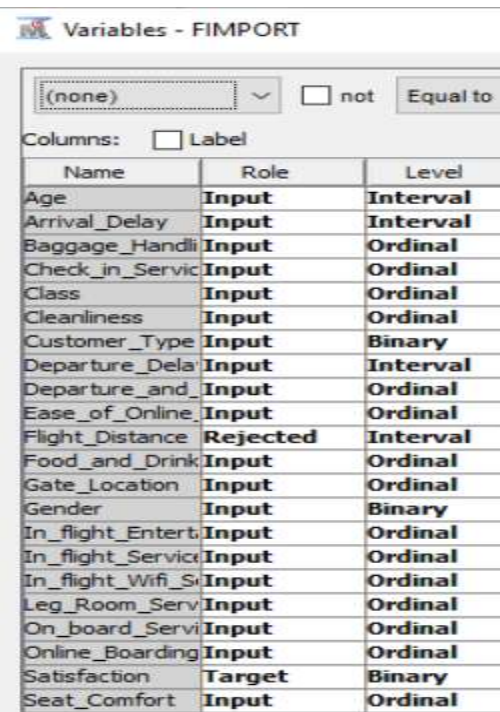
First step of SEMMA framework is to sample the data by creating one or more data tables. Next step is to explore the data by analysing patterns. Modifying the data by selecting, filtering, and transforming the variables to focus the model selection process is next step. The next step in framework is to build a model that predicts a desired outcome. Last step is to assess the data by evaluating the usefulness of our findings. It helps to transform raw data into useful insights, that can provide methods for diverse business problems.



4. Data Exploration

a. Data Types/Masurement Levels:

The dataset had total 22 variables and model role for all variables is set to input by default. The variables like: Gender, Customer type, and Satisfaction are character variables in the data set. As these have only two distinct non-missing levels in the metadata sample, SAS Enterprise Miner identified them as a binary variable. The four variables: Age, Arrival / Departure Delay, Flight Distance are assigned an interval measurement level since they are numeric variables and have more than ten distinct levels. The rest 15 variables have been assigned as nominal measurement level but as they are each numeric variable with more than two but not more than ten distinct non-missing levels in the sample, it was changed to ordinal measurement level.



The screenshot shows the 'Variables - FIMPORT' window in SAS Enterprise Miner. It displays a list of 22 variables with their assigned roles and measurement levels. The variables are listed in a table with columns for Name, Role, and Level. The roles are mostly 'Input', with 'Satisfaction' as 'Target' and 'Flight_Distance' as 'Rejected'. The measurement levels are 'Interval', 'Ordinal', and 'Binary'.

Name	Role	Level
Age	Input	Interval
Arrival_Delay	Input	Interval
Baggage_Handling	Input	Ordinal
Check_in_Service	Input	Ordinal
Class	Input	Ordinal
Cleanliness	Input	Ordinal
Customer_Type	Input	Binary
Departure_Delay	Input	Interval
Departure_and_Arrival_Delay	Input	Ordinal
Ease_of_Online_Booking	Input	Ordinal
Flight_Distance	Rejected	Interval
Food_and_Drink	Input	Ordinal
Gate_Location	Input	Ordinal
Gender	Input	Binary
In_flight_Entertainment	Input	Ordinal
In_flight_Service	Input	Ordinal
In_flight_Wifi_Service	Input	Ordinal
Leg_Room_Service	Input	Ordinal
On_board_Service	Input	Ordinal
Online_Boarding	Input	Ordinal
Satisfaction	Target	Binary
Seat_Comfort	Input	Ordinal

b. Model Roles:

The model role of Satisfaction is set as target variable as customer satisfaction in the airline industry is the outcome that needs to be predicted. Although flight distance had 1.24 percent missing value, it contained so many values considered as outlier which resulted in right-skewness and heavy-tailed distribution. The maximum value of flight distance is 4983 so even when I replaced the missing value using impute node, it was 610.5 which under-represents extreme value. So, I rejected the Flight_distance variable to make the model performance more accurate.

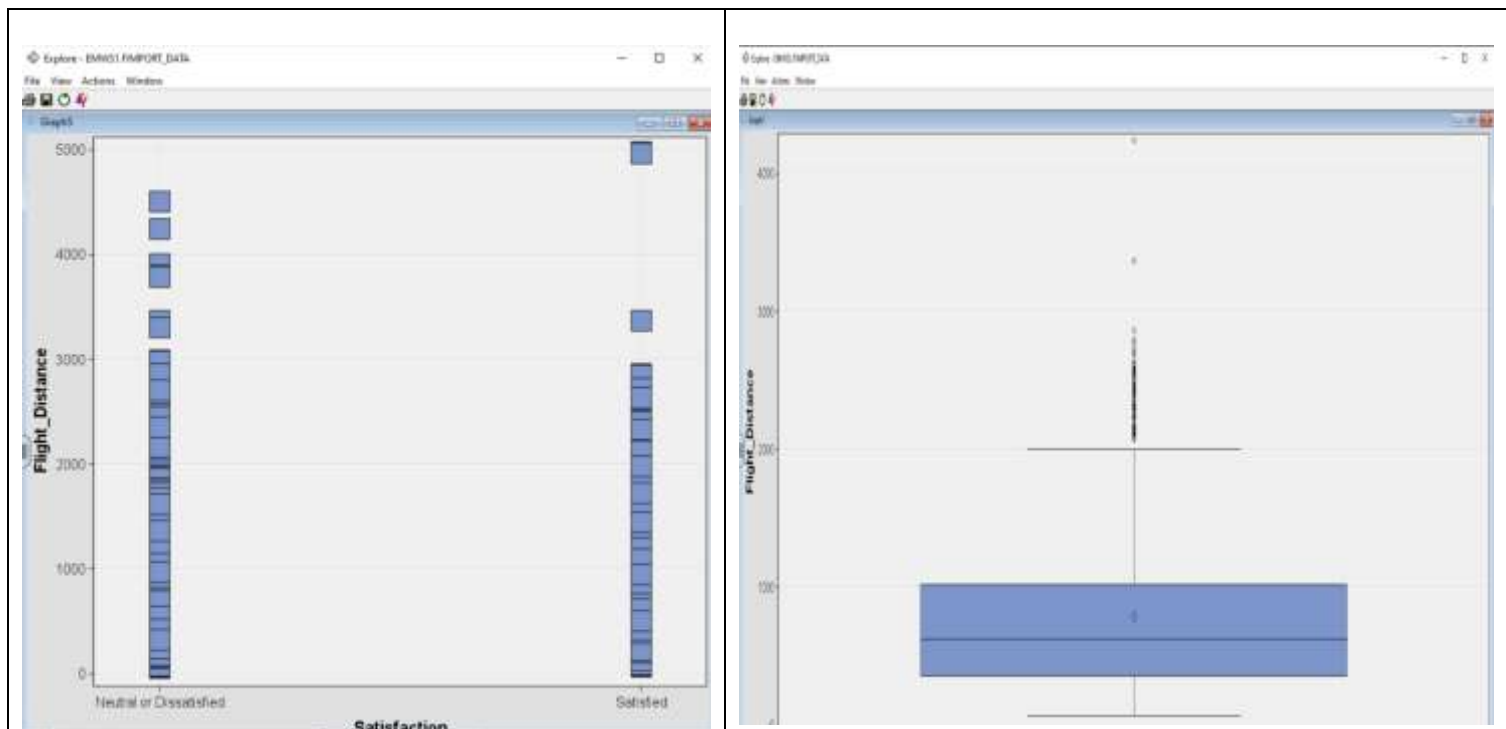


Figure 1: Flight distance and Satisfaction Relation.

c. Dataset Balance:

As we can see in the picture only 33.33% passengers were satisfied leaving 66.67% passengers neutral or dissatisfied. It clearly shows the dataset is imbalance which might bias our model.

Data Role=TRAIN

Data Role	Variable Name	Role	Level	Frequency Count	Percent
TRAIN	Satisfaction	TARGET	Neutral or Dissatisfied	6667	66.67
TRAIN	Satisfaction	TARGET	Satisfied	3333	33.33

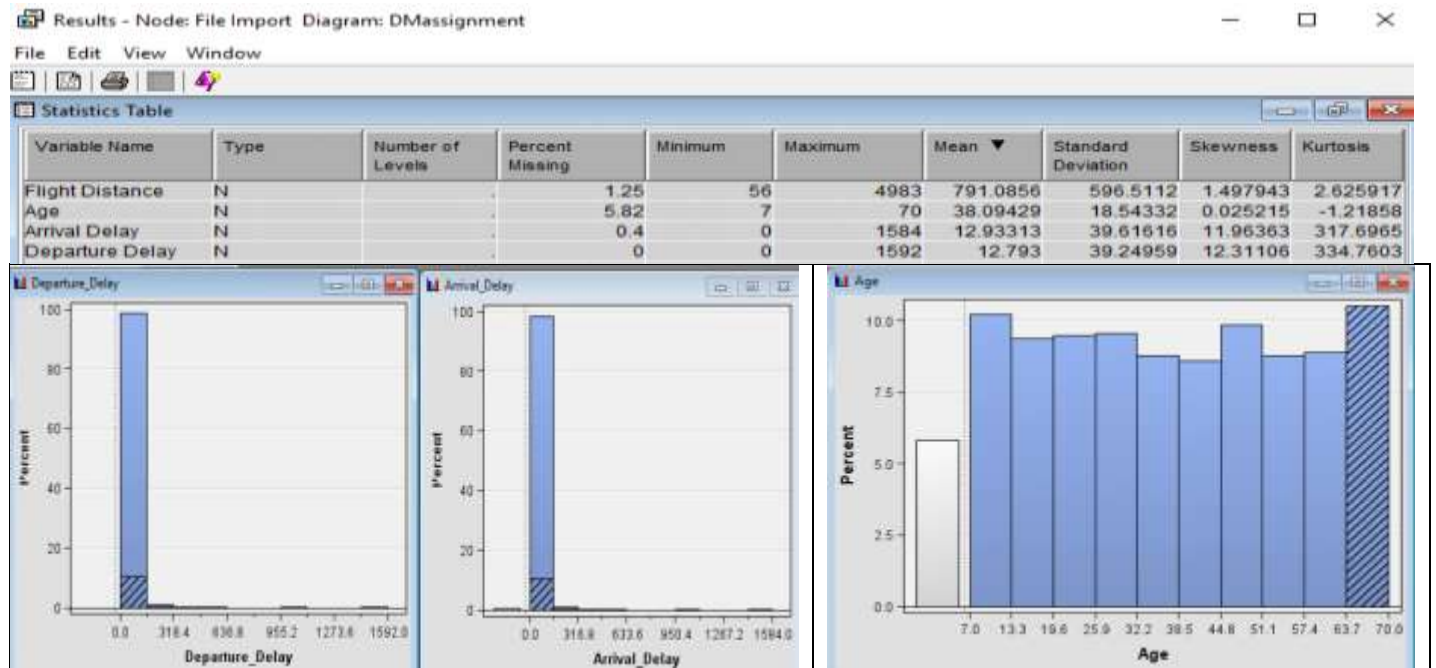
d. Missing Data:

The given dataset had missing values on 3 different variables as shown in the figure:

Results - Node: File Import Diagram: DMassignment			
File Edit View Window			
Statistics Table			
Variable Name	Type	Number of Levels	Percent Missing
Age	N	-	5.82
Flight Distan...	N	-	1.25
Arrival Delay	N	-	0.4

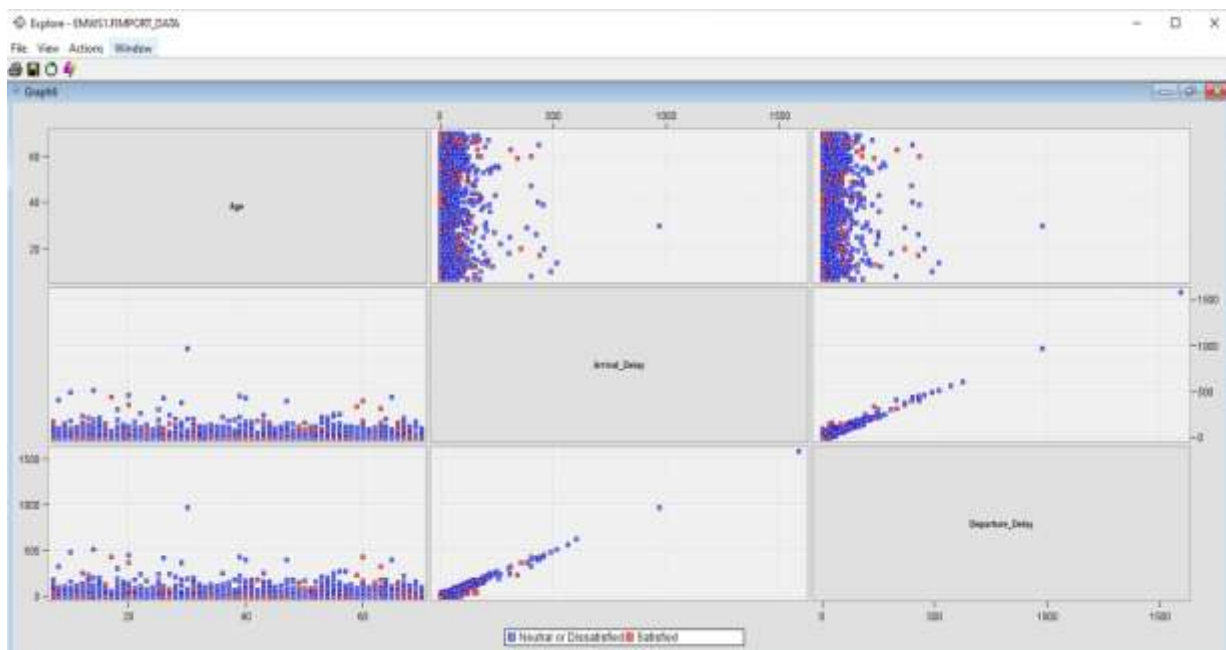
e. Descriptive Statistics:

Both arrival and departure delay showed high right-skewness which can explain the high kurtosis value of 317.69 and 334.76 respectively. Age displayed low skewness and nearly normal distribution.



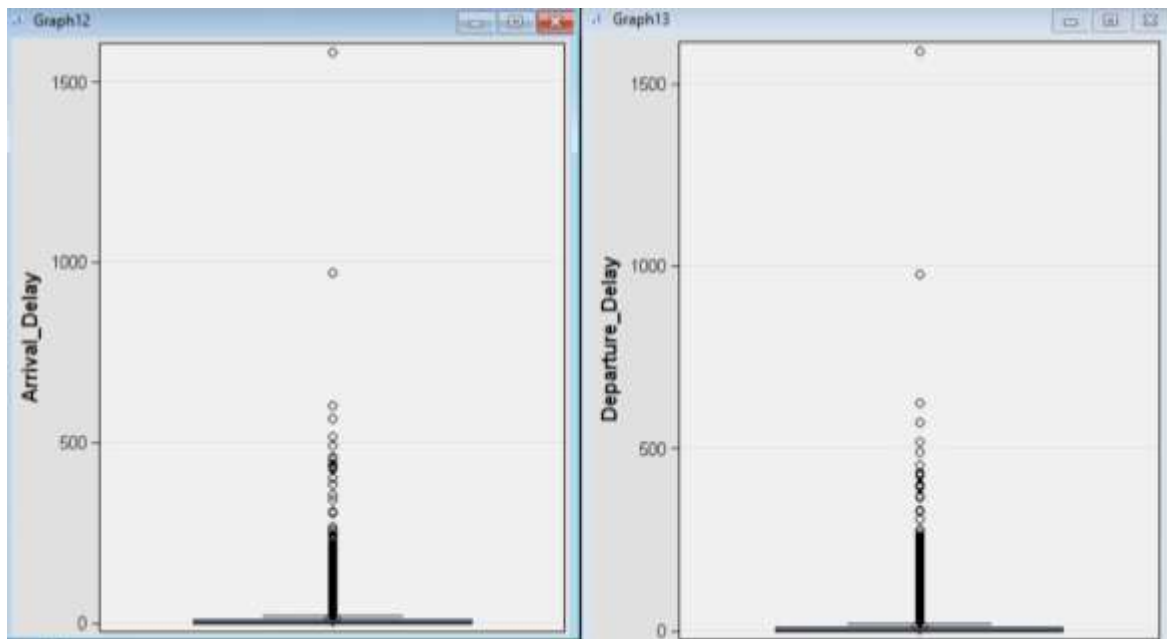
f. Multicollinearity:

We can see from the plot below that both lines are linear so Arrival delay and Departure delay are strongly correlated.



g. Outliers:

From the box plot below here, we can see both arrival and departure delay have extreme values up to 1600 minutes.



h. Sampling Method:

SAS Enterprise Miner choose top sampling method by default. To ensure the target variable maintains the imbalance condition, we change it to Stratified sampling with random seed as 7599.

Property	Value
General	
Node ID	Part
Imported Data	...
Exported Data	...
Notes	...
Train	
Variables	...
Output Type	Data
Partitioning Method	Stratified
Random Seed	7599
Data Set Allocations:	
Training	50.0
Validation	25.0
Test	25.0
Report	
Interval Targets	Yes
Class Targets	Yes
Status	
Create Time	13/03/2025, 16:19
Run ID	8800ccc8-9dfb-41b6-b6!
Last Error	
Last Status	Complete
Last Run Time	27/03/2025, 16:40
Run Duration	0 Hr. 0 Min. 10.62 Sec.
Grid Host	
User-Added Node	No

5. Data partition creation of model sets

I used the Data Partition node's random seed generator to create a random sample of data by entering last 5 digits of my DMU student ID number. This formed a partition in dataset containing:

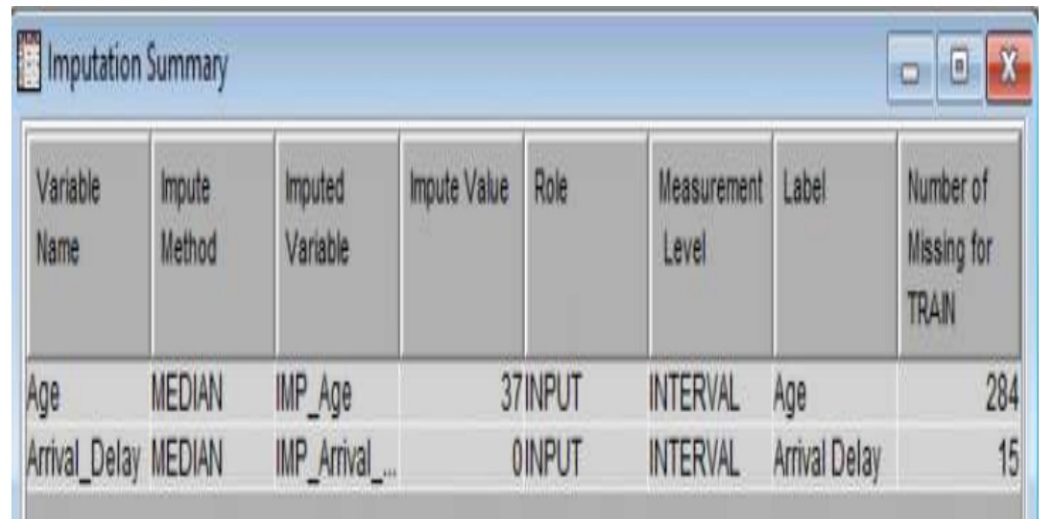
Training = 5000 rows, Validation = 2499 rows, and Test = 2501 rows.

6. Data Modification

a) Data Imputation:

To replace the missing values, I used the median imputation. When I applied the mean imputation, missing value of Age was replaced by 37.69 and Arrival delay was replaced by 13.36.

But to make the passengers satisfied, we need delay of any kind to be 0 so that's why I used median method.



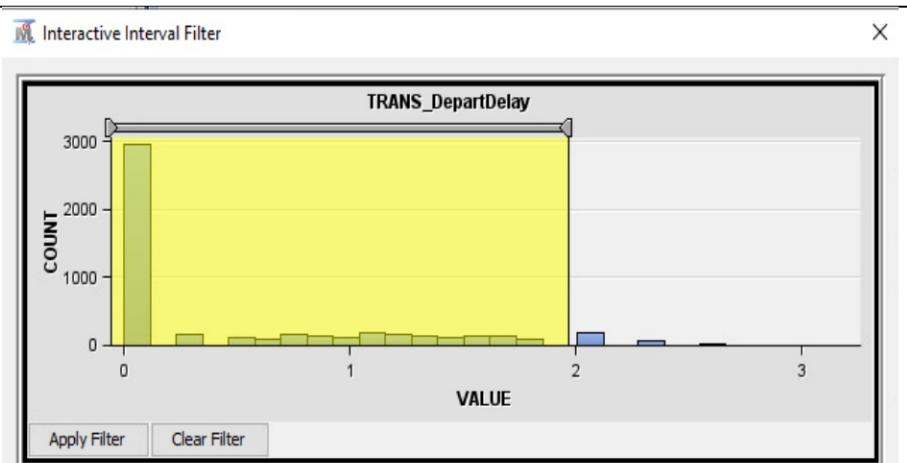
Variable Name	Impute Method	Imputed Variable	Impute Value	Role	Measurement Level	Label	Number of Missing for TRAIN
Age	MEDIAN	IMP_Age	37	INPUT	INTERVAL	Age	284
Arrival_Delay	MEDIAN	IMP_Arrival_...	0	INPUT	INTERVAL	Arrival Delay	15

b) Data Filtration:

I used filter node in three different pipelines with different method containing:

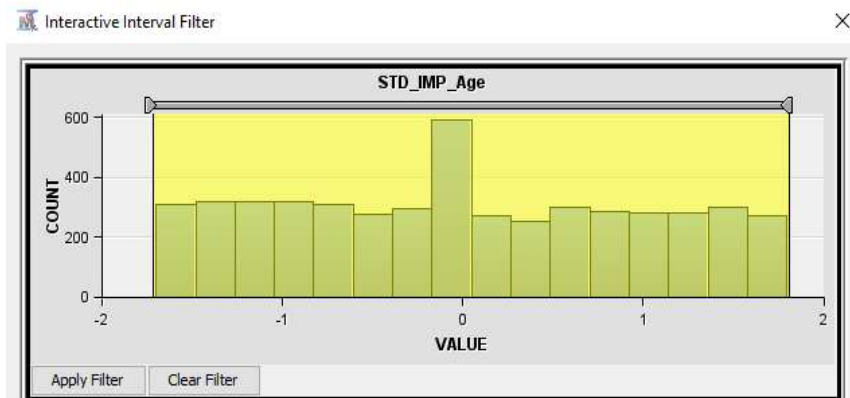
a. Standard Deviation:

Using default filtering method of standard deviation from the mean moved skewed distribution.



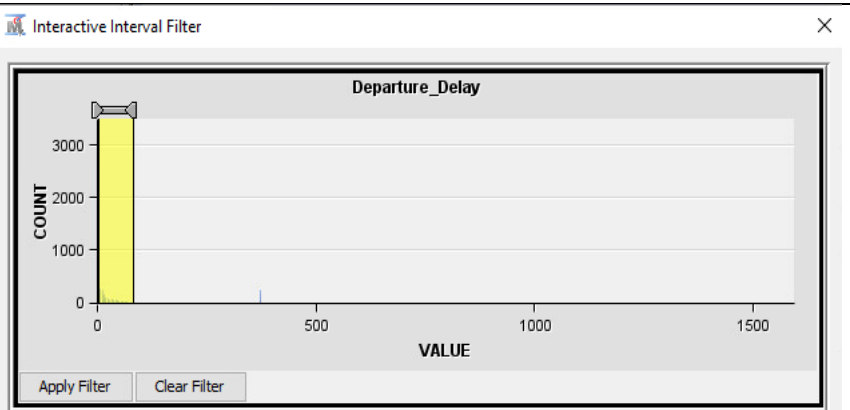
b. User – Specified Limits:

Using user specified limits, this removed extreme outliers which improved data quality.



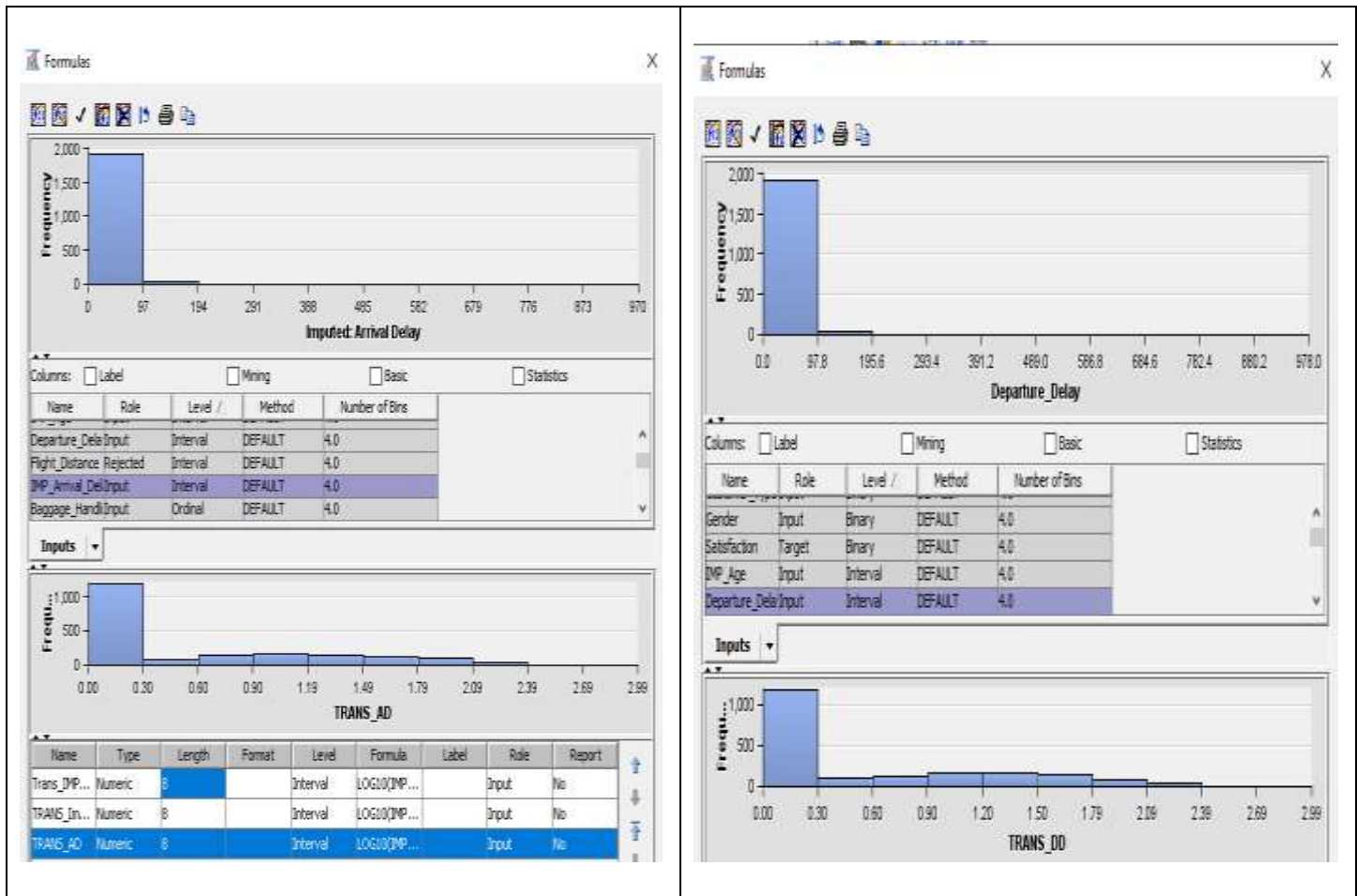
c. Mean Absolute Deviation:

Using mean absolute deviation as filtering method, it filtered the variables by removing extreme outliers.



c) Data Transformation:

Both arrival delay and departure delay were highly right skewed so I used log transformation to make it more normally distributed. Even though the plot was not likely normally distributed but it reduced the skewness and minimized outlier impact.



7. Data Modelling

I. Neural Network

i. Development of models

Table 3: Neural Network Model Variations and Their Rationale

Model Variation	Reason
Default-Neural-Network-Principal-components (NN-1)	To see the impact before applying any data preprocessing methods, we can establish a baseline model while compressing correlated variables.
Imp-Filter-VSchi2-Neural-Network (NN-2)	With limited 3 hidden nodes which can optimize efficiency and using Chi2 variable-selection to filter irrelevant variables.
Imp-Transform-Filter-VSchi2-Neural-Network (NN-3)	With hidden node set to 5 and maximum iteration set to 25, it was chosen to propagate errors backward. This improve training pace while reducing noise and preventing overfitting with the use of Chi2 selection.
Imp-Transform-Filter-VSr2-Neural-Network (NN-4)	Used back propagation to generate stable outcomes and to prevent overfitting, used 10 hidden units with 25 iterations.

ii. Model performance

Table 4: Neural network model performance

Model variation	ROC Index	Cu. Lift	Scope %	True – %	False – %	True + %	False + %
1. NN-1	0.959	2.18	45%	90.58%	14.06%	85.93%	9.41%
2. NN-2	0.974	2.1	45%	90.4%	8.2%	91.8%	9.6%
3. NN-3	0.83	2.15	35%	90.9%	37.87%	62.13%	9.1%
4. NN-4	0.923	2.2	45%	86.56%	13.83%	86.17%	13.44%

291	Event Classification Table								
292	Model Selection based on Valid: Misclassification Rate (_VMISC_)								
293									
294	Model		Data			False	True	False	True
295	Node	Model Description	Role	Target	Target Label	Negative	Negative	Positive	Positive
296									
297	Neural	Default-NN	TRAIN	Satisfaction	Satisfaction	177	3004	329	1490
298	Neural	Default-NN	VALIDATE	Satisfaction	Satisfaction	117	1510	157	715
299	Neural2	Imp-F-VSchi2-NN	TRAIN	Satisfaction	Satisfaction	83	3002	257	1551
300	Neural2	Imp-F-VSchi2-NN	VALIDATE	Satisfaction	Satisfaction	68	1507	160	764
301	Neural3	Imp-Tr-Fr-VSchi2-NN	TRAIN	Satisfaction	Satisfaction	587	3073	239	1056
302	Neural3	Imp-Tr-Fr-VSchi2-NN	VALIDATE	Satisfaction	Satisfaction	315	1516	151	517
303	Neural4	Imp-Tr-F-VSr2-NN	TRAIN	Satisfaction	Satisfaction	196	2885	433	1448
304	Neural4	Imp-Tr-F-VSr2-NN	VALIDATE	Satisfaction	Satisfaction	115	1443	224	717

From this table we are getting numbers of true positive (TP), true negative (TN) and false positive (FP), false negative (FN). To calculate each term in percentage we have formula as given below:

$TP \% = \frac{TP}{TP + FN} * 100 \%$	$TN \% = \frac{TN}{TN + FP} * 100 \%$
$FN \% = 100 \% - TP \%$	$FP \% = 100 \% - TN \%$

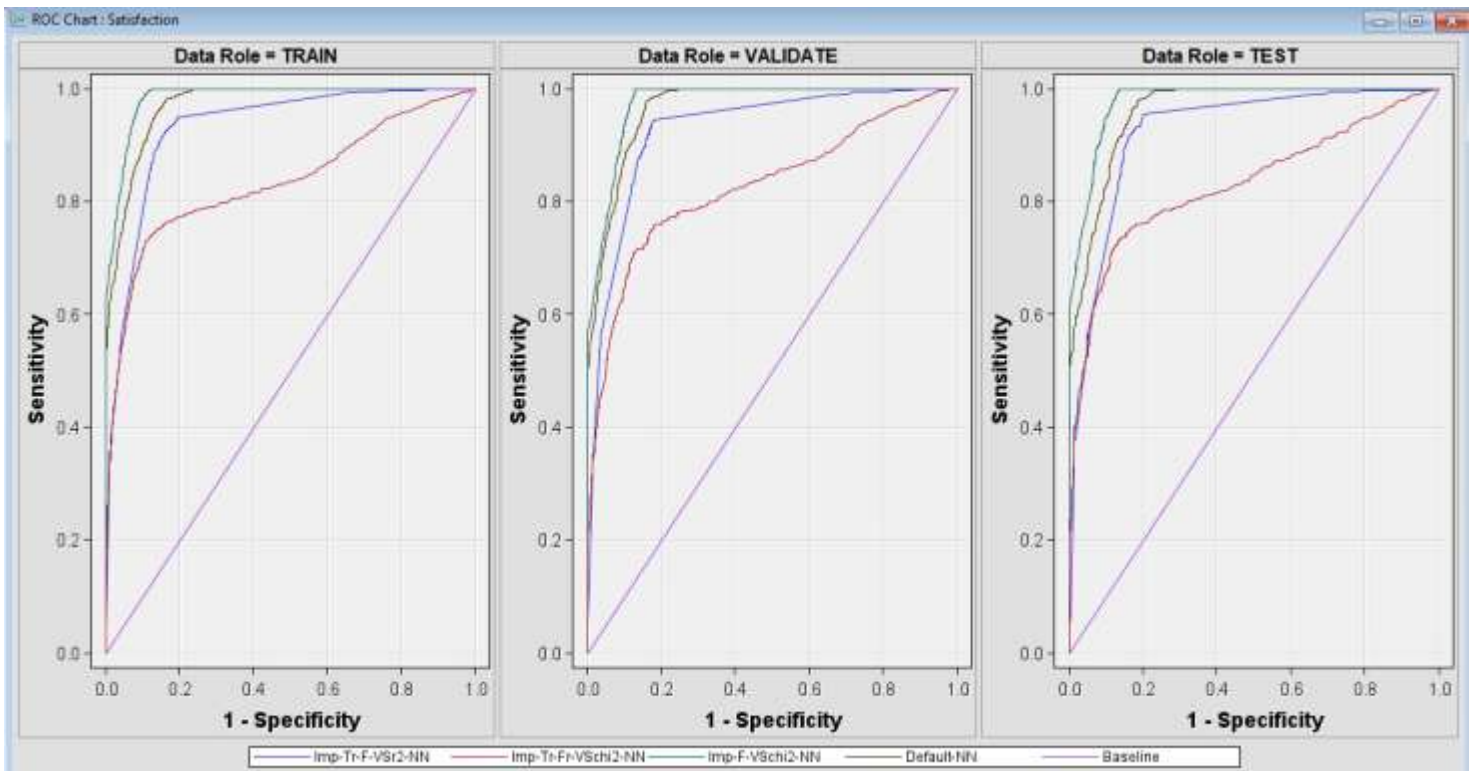


Figure 2: Neural Network ROC Chart

This ROC curve above shows that Neural Network with impute, filter and variable selection with χ^2 model outperforms all other models having the highest roc index of 0.974.

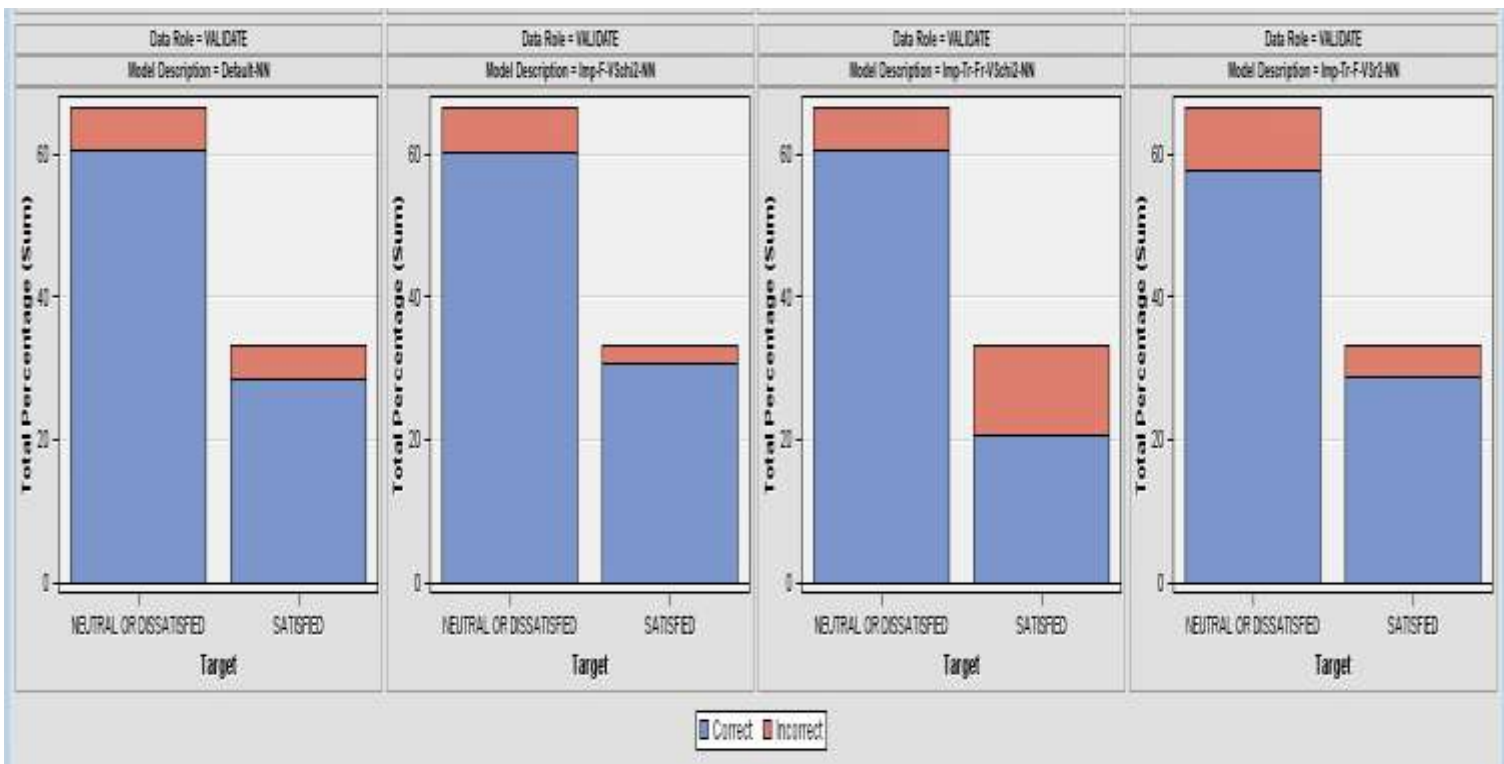


Figure 3: NN - Classification Chart.

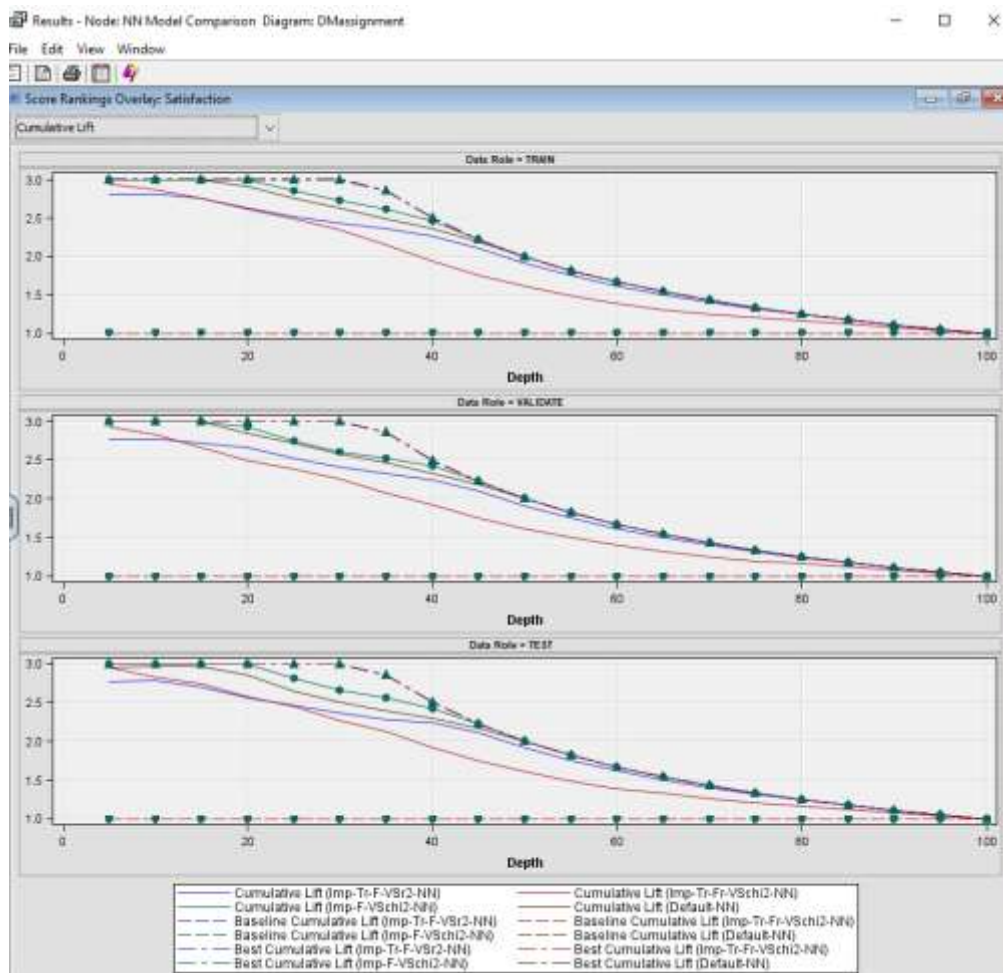


Figure 4: NN - Cumulative Lift.

I have made the comparisons of my model's output with baseline model and best possible model. Model that had reliable cumulative lift curve and steady decrease to baseline was Neural Network 2. So, it was considered the best model.

iii. Overfitting analysis:

The best model NN-2 achieved the lowest error (6.95 % train and 9.12 % validation) confirming our result. As all models showed small gaps (< 3%), it shows there was no overfitting.

II. Decision Tree

a. Development of models

Table 5: Decision Tree Model Variations and Their Rationale

Model variation	Reason
Tr-Entropy-Binary-Split-Decision-Tree (DT-1)	Uses Entropy for target criterion to balance class distributions.
Filter-Gini-Binary-Split-Decision Tree (DT-2)	Uses Gini-entropy which is optimal to compute large datasets.
VS-Entropy-Multiway-Split-Decision-Tree (DT-3)	Uses entropy-based multiway splits which captures complex category patterns.
Default Gini-Multiway-Split-Decision-Tree (DT-4)	Uses Gini-based multiway splits which provides a baseline with faster training.

b. Model performance

Table 6: Decision tree model performance

Model variation	ROC Index	Cu. Lift	Scope %	True – %	False – %	True + %	False + %
DT-1	0.949	2.75	50%	87.2%	0%	100%	12.8%
DT-2	0.949	2.75	50%	87.2%	0%	100%	12.8%
DT-3	0.976	3.003	50%	90.82%	6.8%	93.2%	9.18%
DT-4	0.976	3.003	50%	90.82%	6.8%	93.2%	9.18%

Event Classification Table

Model Selection based on Valid: Misclassification Rate (_VMISC_)

Model Node	Model Description	Data Role	Target	Target Label	False Negative	True Negative	False Positive	True Positive
Tree2	DTGini-Binary\$split-pruned	TRAIN	Satisfaction	Satisfaction	0	2881	417	1636
Tree2	DTGini-Binary\$split-pruned	VALIDATE	Satisfaction	Satisfaction	0	1455	212	832
Tree3	DTEntropy-Multiway\$split	TRAIN	Satisfaction	Satisfaction	83	3075	258	1584
Tree3	DTEntropy-Multiway\$split	VALIDATE	Satisfaction	Satisfaction	56	1514	153	776
Tree4	DTGini-Multiway\$split-Pruned	TRAIN	Satisfaction	Satisfaction	83	3075	258	1584
Tree4	DTGini-Multiway\$split-Pruned	VALIDATE	Satisfaction	Satisfaction	56	1514	153	776
Tree	DTEntropy-Binary\$split	TRAIN	Satisfaction	Satisfaction	0	2913	420	1667
Tree	DTEntropy-Binary\$split	VALIDATE	Satisfaction	Satisfaction	0	1455	212	832

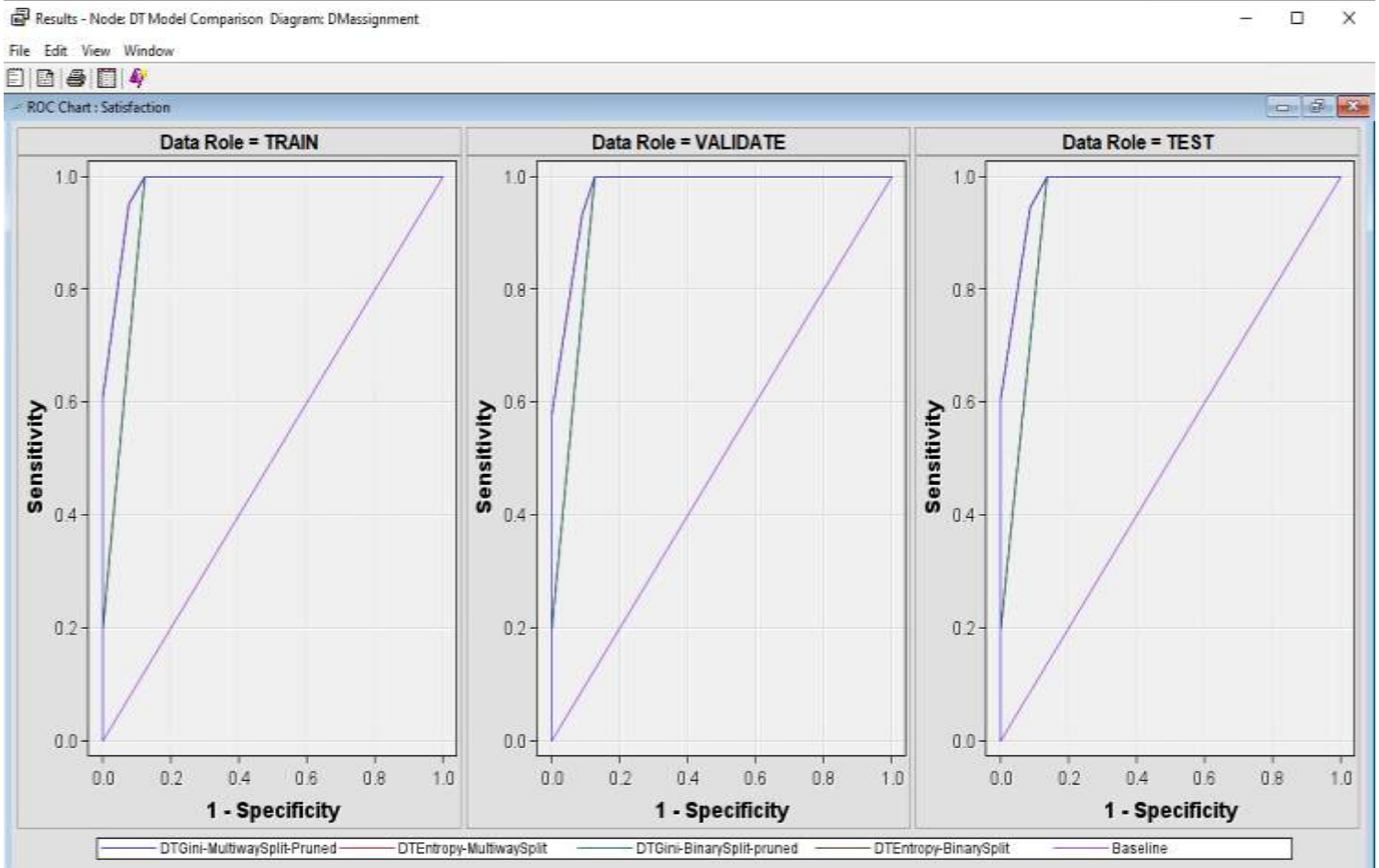


Figure 5- Decision Tree ROC Chart.

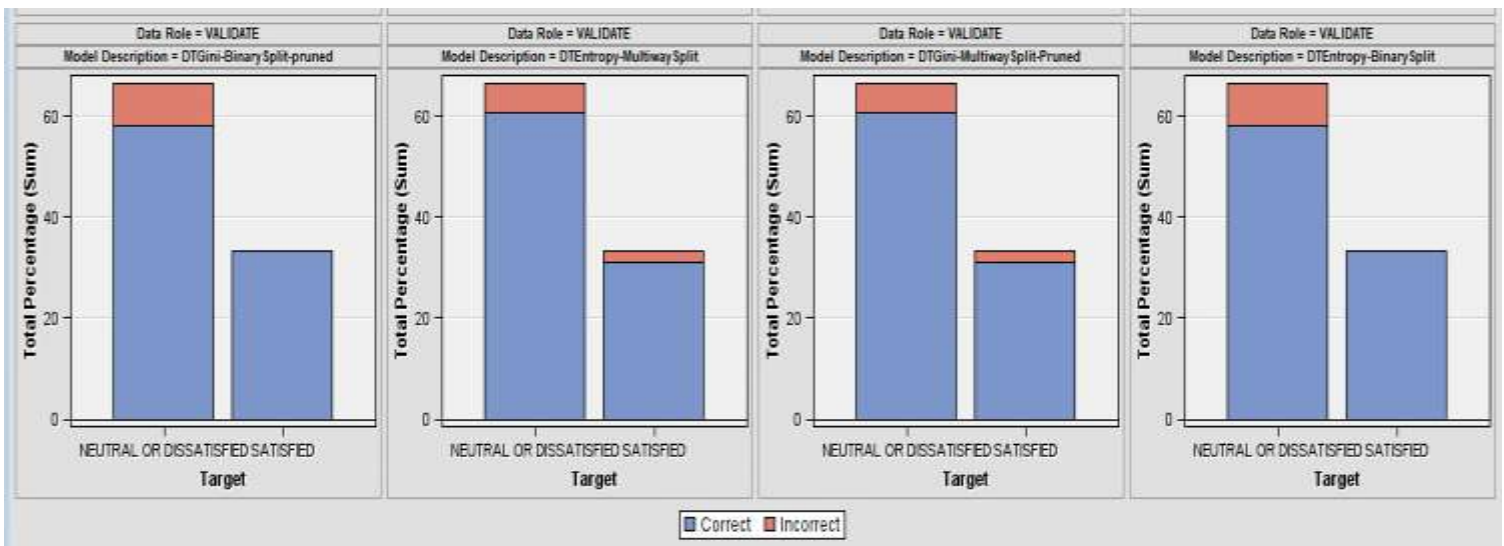


Figure 6: DT - Classification Chart.

Based on ROC index, Decision-Tree3 outperforms all the models. The cumulative-lift chart below shows the most likely positive cases at the top of the list by the same model.

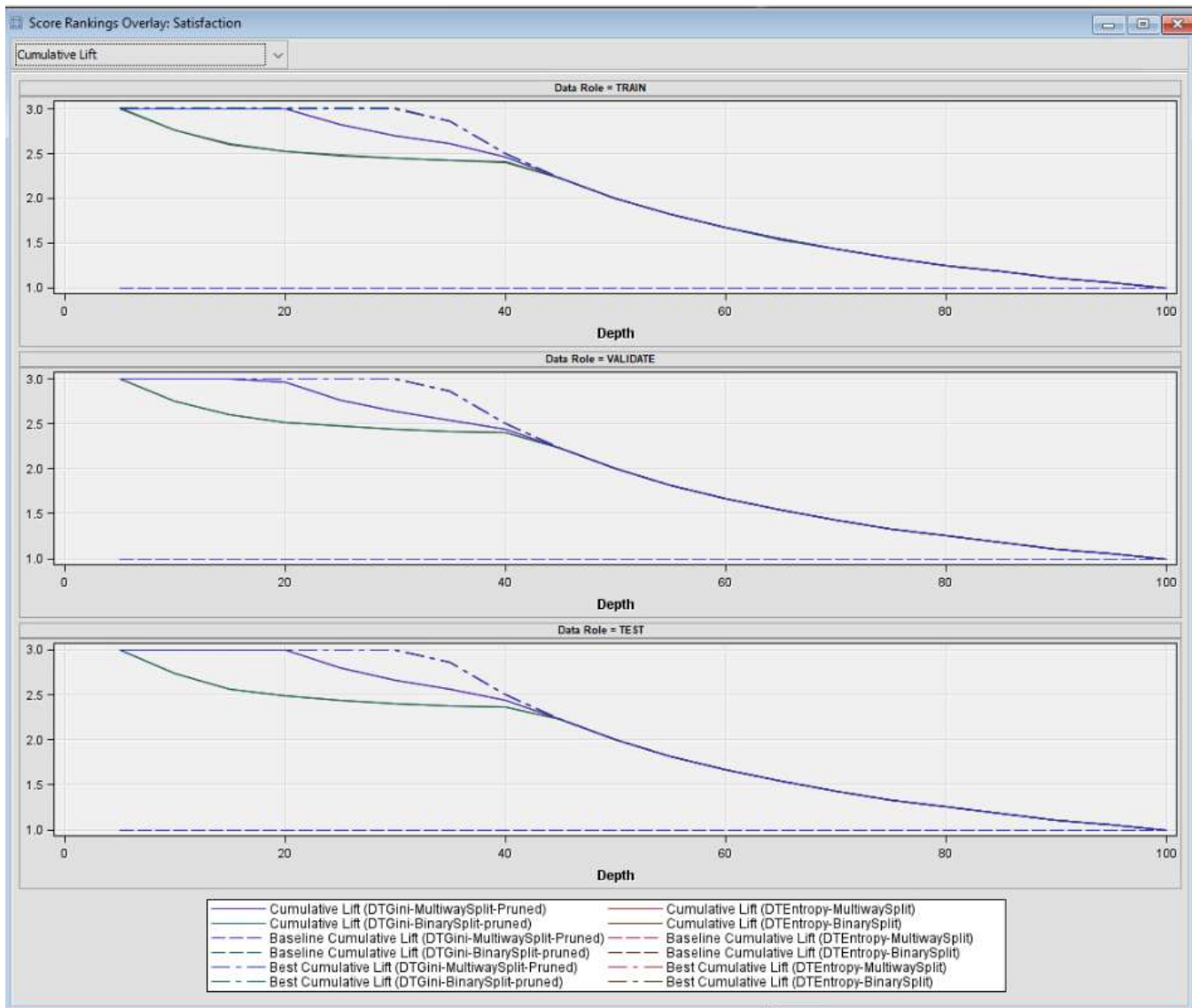


Figure 7: DT - Cumulative Lift.

c. Critical path of best model

Decision Tree with entropy and multiway-split (DT-3) model was selected as the best model due to its lowest classification rate (6.8% training, 8.3% validation).

d. Target path of interest

FileImport → DataPartiton → VariableSelection → DecisionTree → ModelComparison

e. Overfitting analysis

As all the trained models had small gaps (< 3%), it shows there was no overfitting.

8. Analysis of the best model

Table 7: Summary results of the best performing models

Model	ROC-Index	Cu.-Lift	Scope%	True(-)%	False(-)%	True(+)%	False(+)%
NN- 2	0.974	2.1	45%	90.4%	8.2%	91.8%	9.6%
DT - 3	0.976	3.003	50%	90.82%	6.8%	93.2%	9.18%

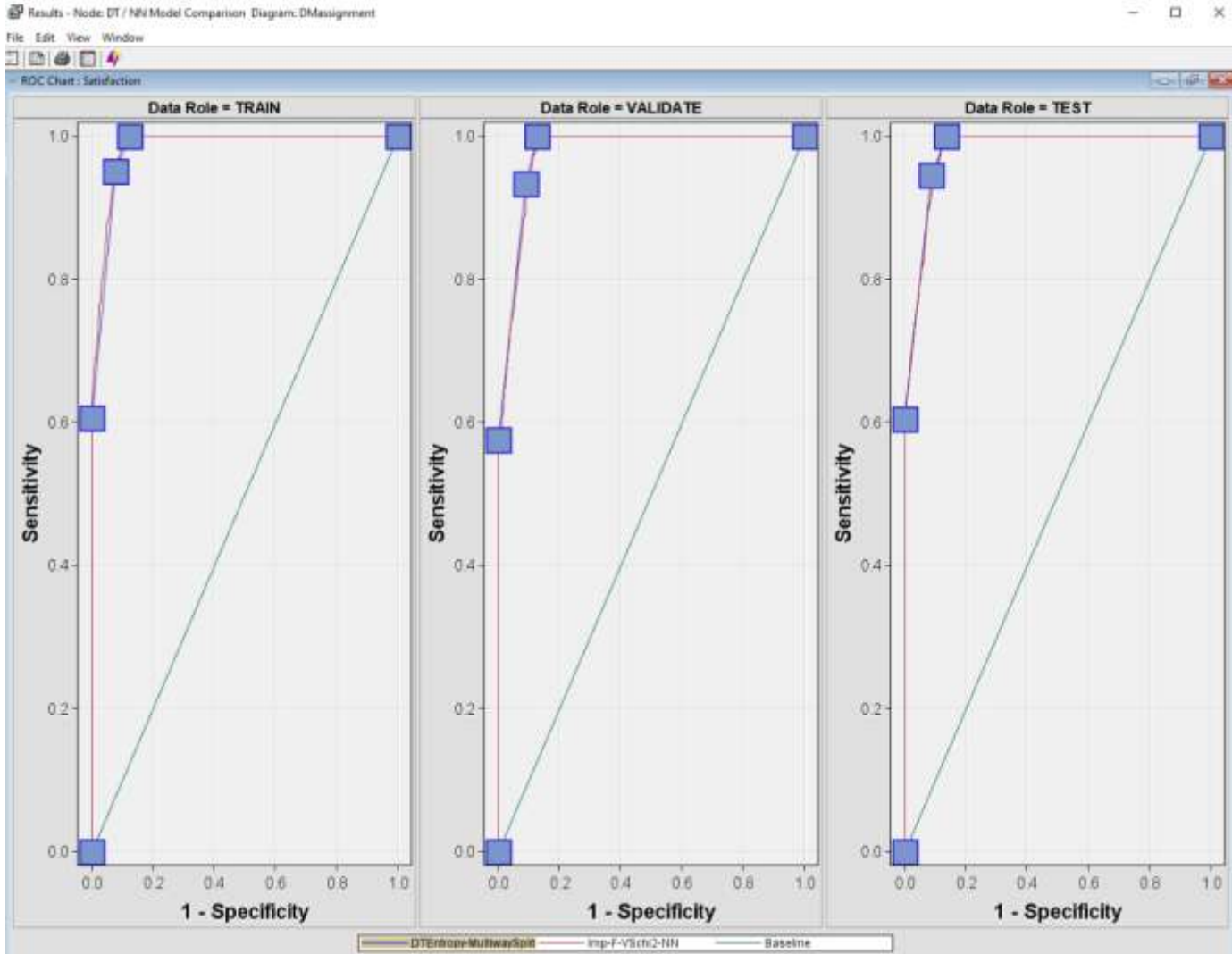


Figure 8: DT and NN ROC Chart.

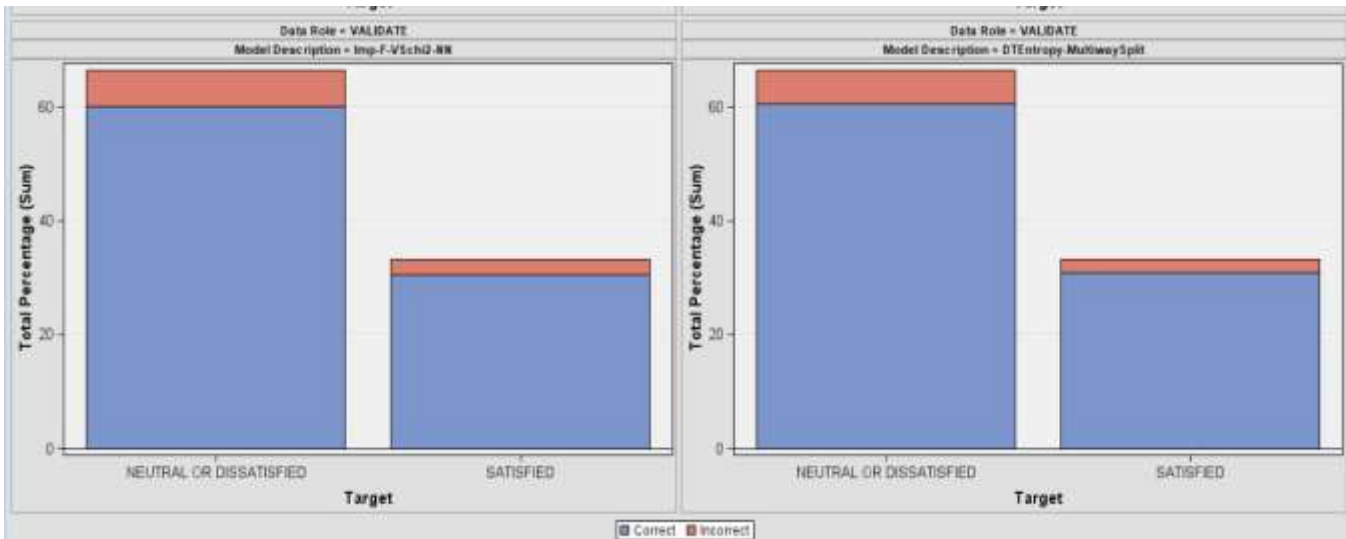
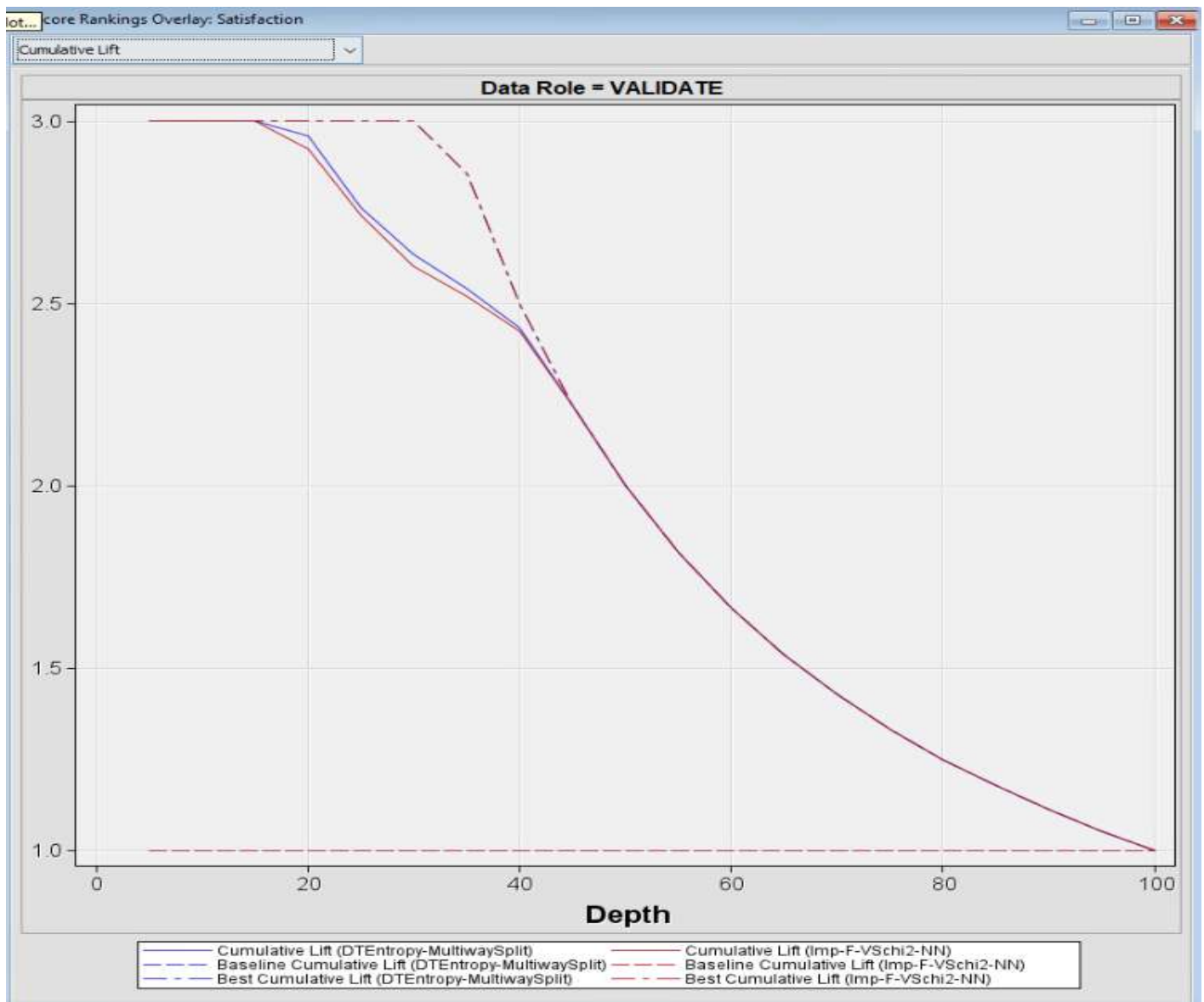


Figure 9: DT and NN Classification Chart and Cumulative lift.



9. Breadth of areas of application and research in data mining

Our module only covered basic techniques of modeling. As predictive modeling is rapidly evolving in today's market, we can make use of this in many research areas. It uses the information gathered from data in order to answer questions about future scenarios. One area that profited the most with the use of data mining is Business sector. The exploration of data mining for businesses still continues to expand as e-commerce and e-marketing mainstreams in the retail section. As businesses grow every day, we can see the development of more application-specific data mining tools such as increasing user interaction. Another field that can harness data mining tools are biomedical informatics. We can use data mining to analyse biomedical data and support analysis in areas such as: cancer detection with use of IoT. The information related to lung cancer taken from IoT devices are utilized for knowing and managing difficult environments. In these environments, challenge will be the time to process huge amounts of data to deliver where data mining can be fit.

10. Conclusion and recommendations

This project identifies the customer dissatisfaction of an airline industry. We found the major factors that affected the experiences was in-flight services and flight delays while excluding the irrelevant factors like flight distance. Decision Tree model (DT-3) gave us the better performance with less error of 6.8% so I recommend using this model for prediction.

11. My Reflections on the process

I learned the process of data imputation, filtering, transformation and also got to know that how selection of model can impact the predictive accuracy. From this project, I found that airline businesses need to prioritize on key factors that effect their customer. Most importantly, I discovered that best model balances accuracy and business impact.

12. References

(Han, 2012)

(Pandit, 2021)

(Pradhan Kanchan, 2020)

13. Appendix

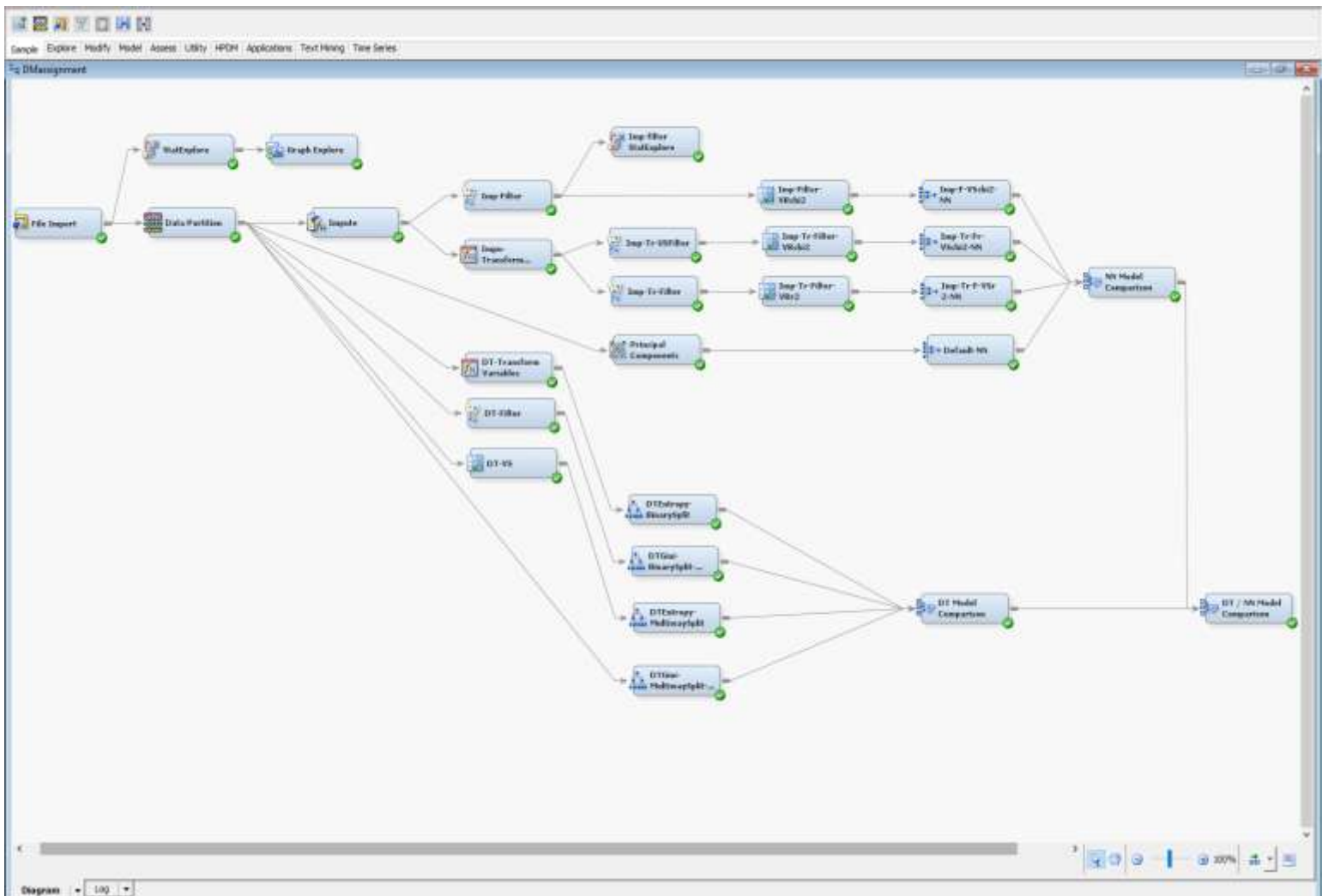


Figure 10: Workflow Diagram.